

DUGGINALA AJAY KUMAR

+91 7995549621 ajaykumardugginala@gmail.com Hyderabad, Telangana, India
linkedin.com/in/ajay-kumar-dugginala github.com/ajaykumar-dugginala

RESEARCH INTEREST

Computer Science undergraduate with strong foundations in algorithms, data structures, and software engineering, actively developing expertise in Machine Learning and Natural Language Processing. Interested in research at the intersection of AI systems and real-world problem solving. Eager to contribute to and learn from cutting-edge research at Google, with the goal of advancing intelligent systems that scale to billions of users.

EDUCATION

MLR Institute of Technology

Hyderabad, Telangana

Bachelor of Technology in Computer Science & Engineering — **CGPA: 8.5 / 10.0**

2023 – 2027

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Linear Algebra, Probability & Statistics
- Currently exploring: Machine Learning, Natural Language Processing, Deep Learning fundamentals

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, SQL

ML / AI Libraries: NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, NLTK (learning: PyTorch, TensorFlow)

Web Technologies: React.js, HTML5, CSS3, REST APIs

Databases: MySQL, Relational Schema Design, Query Optimization, ACID Transactions

Core CS: Data Structures, Algorithms, OOP, DBMS, Operating Systems, System Design

Tools & Platforms: Git, GitHub, VS Code, Linux, Postman, Jupyter Notebook, Google Colab

Problem Solving: Competitive Programming, Dynamic Programming, Graph Theory, Recursion

PROJECTS

Sentiment Analysis on Product Reviews | *Python, NLTK, scikit-learn, Pandas, TF-IDF*

2025

- Built an end-to-end NLP pipeline to classify Amazon product reviews as positive, negative, or neutral using TF-IDF vectorization and Logistic Regression, achieving **85%+ accuracy** on a 50,000-record dataset.
- Performed data preprocessing steps including tokenization, stopwords removal, lemmatization, and handling class imbalance using SMOTE — directly applying NLP research concepts to real-world data.
- Compared performance of Naive Bayes, SVM, and Logistic Regression classifiers; analyzed confusion matrices and F1 scores to identify the optimal model for deployment.
- Visualized word frequency distributions and sentiment trends across product categories using Matplotlib and WordCloud, enabling data-driven insights for hypothetical business use.

Spam Email Classifier | *Python, scikit-learn, Pandas, NumPy, Bag-of-Words*

2025

- Designed a binary text classification system to detect spam emails using Bag-of-Words and Multinomial Naive Bayes, achieving **97% precision** on the UCI Spambase benchmark dataset.
- Conducted exploratory data analysis (EDA) to identify discriminative features; applied feature selection using chi-squared tests to reduce dimensionality and improve model generalization.
- Evaluated model robustness via k-fold cross-validation and ROC-AUC analysis, documenting methodology and findings in a structured experimental report — mirroring academic research practices.

Data Management System | *Python, MySQL, RDBMS, Query Optimization*

2024

- Designed and implemented a full-featured CRUD application with a normalized relational database schema (3NF), eliminating data redundancy and ensuring referential integrity.
- Optimized complex SQL queries using indexed lookups and JOIN operations, achieving sub-100ms response times on datasets exceeding 10,000+ records.
- Developed parameterized queries to prevent SQL injection vulnerabilities, following OWASP secure coding guidelines, and implemented ACID-compliant transaction management.

Algorithmic Console Application | *C++, Java, Data Structures, Algorithm Design*

2024

- Engineered a modular command-line application demonstrating advanced algorithmic paradigms: divide-and-conquer, dynamic programming, and backtracking recursion.
- Implemented core data structures (stacks, queues, linked lists, binary trees) from scratch; achieved $O(n \log n)$ average-case time complexity across sorting and searching modules.

RESEARCH EXPOSURE & SELF-DIRECTED LEARNING

Independent ML/NLP Study

Online / Self-Directed

Focused coursework in Machine Learning and Natural Language Processing

2024 – Present

- Studied core ML theory: supervised/unsupervised learning, bias-variance tradeoff, regularization, ensemble methods — drawing from Andrew Ng’s ML Specialization and fast.ai resources.
- Read and summarized foundational NLP papers including *Attention Is All You Need* (Vaswani et al.) and *BERT: Pre-training of Deep Bidirectional Transformers* (Devlin et al.) to build intuition for transformer-based architectures.
- Completed hands-on experiments in Google Colab applying learned concepts to publicly available datasets; documented findings in structured lab-style notebooks.

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- **CodeChef Silver Badge** — Consistent high performance across multiple algorithmic contest rounds, demonstrating strong problem-solving speed and accuracy under pressure.
- **200+ DSA Problems Solved** — Covering Arrays, Strings, Recursion, Sorting, Searching, Trees, Graphs, and Dynamic Programming across platforms including CodeChef and LeetCode.
- Strong proficiency in time & space complexity analysis; consistently optimizes from brute-force $O(n^2)$ to optimal $O(n \log n)$ solutions.
- **CGPA: 8.5/10.0** at MLR Institute of Technology — maintained through a rigorous CS curriculum while simultaneously pursuing independent research and ML study.

CORE COMPETENCIES

Research Curiosity · Analytical Thinking · Problem Decomposition · Experimental Methodology · Team Collaboration · Attention to Detail · Fast Learner · Scientific Communication · Time Management